

# ALBERTO LARRAURI

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## EMPLOYMENT AND EDUCATION

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**Assistant Professor (Profesor Ayudante Doctor)** *September 2025 -*  
Universidad de Zaragoza.

**Post Doctoral Research Associate** *July 2025 - September 2025*  
Durham University.  
Funded by the UKRI grant *Promise Constraint Satisfaction Problem: Structure and Complexity*.

**Senior Research Associate** *July 2023 - July 2025*  
University of Oxford.  
Funded by the UKRI (formelry ERC) grant *New Approaches to Approximability of Satisfiable Problems*.

**Junior Research Fellow** *October 2024 -*  
Kellogg College, University of Oxford.

**Research assistant** *October 2020 - December 2022*  
Technical University of Graz.

**PhD in Applied Mathematics** *September 2019 - March 2023*  
Universitat Politècnica de Catalunya.  
Funded by the ERC grant *A Unified Theory of Algorithmic Relaxations*.  
Awarded the Excelent Cum Laude grade.

**Master's degree in Advanced Mathematics and Mathematical Engineering**  
*September 2018 - July 2019*  
Universitat Politècnica de Catalunya.

**Bachelor's degree in Mathematics** *September 2014 - July 2018*  
Universidad de Zaragoza.

## RESEARCH INTERESTS

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Constraint satisfaction problems, algorithms and complexity, combinatorics, finite model theory, random structures.

## PEER-REVIEWED PUBLICATIONS

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*Ineffectiveness for Search and Undecidability of PCSP Meta-Problems*, A. Larrauri. FOCS, 2025.

*Optimal Inapproximability of Promise Equations over Finite Groups*, S. Butti, A. Larrauri, S. Živný. ICALP, 2025.

*Convergence Laws for Extensions of First-Order Logic with Averaging*, S. Adam-Day, M. Benedikt, A. Larrauri. LICS, 2025.

*Synthesis of Controllers for Continuous Blackbox Systems*, B. Maderbacher, F. Windisch, A. Larrauri, and R. Bloem. International Conference on Verification, Model Checking, and Abstract Interpretation, 2025.

*Solving promise equations over monoids and groups*, A. Larrauri, S. Živný. ICALP, 2024. A journal version was published in ACM Transactions on Computational Logic, 2024.

*Limiting probabilities of first order properties of random sparse graphs and hypergraphs*, A. Larrauri, T. Müller, M. Noy. Random Structures & Algorithms, 2022.

*Industry Paper: Surrogate Models for Testing Analog Designs under Limited Budget a Bandgap Case Study*, R. Bloem, A. Larrauri, R. Lengfeldner, C. Mateis, D. Ničković & B. Ziegler. Emebedded Systems Week (ESWEEK), 2022.

*Probabilities of first order sentences on sparse random relational structures: An application to definability on random CNF formulas*, A. Larrauri. Journal of Logic and Computation, 2021.

## PREPRINTS AND ARTICLES IN PREPARATION

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*Equations over Finite Monoids with Infinite Promises*, A. Larrauri, A. Mottet, S. Živný. Accepted at Transactions on Computational Logic. Preprint available as arXiv:2502.06762.

*Limiting Probabilities of First Order Properties of Sparse Graphs with Given Degree Sequences*, A. Larrauri, G. Perarnau. Preprint available as arXiv:2405.14836. Under submission at Random Structures and Algorithms.

*Minimization and Synthesis in Sequential Compositions of Mealy Machines: Revisited*, A. Larrauri, R. Bloem. Preprint available as arXiv:2105.10292.

*Conformance Testing of Mealy Machines Under Input Restrictions*, A. Larrauri, R. Bloem. Preprint available as arXiv:2206.07441.

*Preservation Theorems on Random Graphs*, A. Larrauri, in preparation.

## THESES

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*First order logic of random sparse structures*,  
PhD Thesis, Universitat Politècnica de Catalunya, 2023.

*First order logic of sparse hyper-graphs*,  
MSc Thesis, Universitat Politècnica de Catalunya, 2020.

## TEACHING

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Algebra, main lecturer, Computer Engineering BSc, Escuela Politcnica de Teruel, Sept. 2025 - Jan. 2026.

Algebra, main lecturer, Electronic Engineering BSc, Escuela Politcnica de Teruel, Sept. 2025 - Jan. 2026.

Calculus, main lecturer, Computer Engineering BSc, Escuela Politcnica de Teruel, Sept. 2025 - Jan. 2026.

Calculus, main lecturer, Electronic Engineering BSc, Escuela Politcnica de Teruel, Sept. 2025 - Jan. 2026.

Quantum Information, class tutor, Computer Science BSc (optional) / MSc in Advanced Computer Science (optional), University of Oxford, Hilary Term 2025.

Probabilistic Combinatorics, class tutor, Mathematics BSc (optional), University of Oxford, Hilary Term 2024.

Combinatorial Optimization, class tutor, Computer Science BSc (optional), University of Oxford, Michaelmas Term 2023.

Logic and Computability, supporting role. Computer Science BSc (mandatory), TU Graz. Summer Semester 2022.

Model Checking, supporting role. Computer Science MSc (optional), TU Graz. Summer Semester 2021.

Problem Workshop. 1st and 2nd Semester Mathematics BSc, Universitat Politècnica de Catalunya. Sep. 2019 - Jul. 2020.

## PARTICIPATION IN RESEARCH PROJECTS

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Supported in the past by:

- The UKRI project *Promise Constraint Satisfaction Problem: Structure and Complexity* (EP/X033201/1).
- The UKRI (formerly ERC) project *New Approaches to Approximability of Satisfiable Problems* (EP/X024431/1).
- The ERC project *A Unified Theory of Algorithmic Relaxations* (H2020-648276),
- The FFG project ADVANCED (874044), and
- The FFG project FATE (894789).

Part of the *Working Team* at:

- COCOA: COntemporary COmbinatorics and its Applications, (PID2023-147202NB-I00) funded by the Spanish Ministry for Science and Innovation.
- *Combinatorics: new trends and real-world applications*, (PID2020-113082GB-I00) funded by the Spanish Ministry for Science and Innovation.
- *Geometric, algebraic and probabilistic combinatorics* (MTM2017-82166-P) funded by the Spanish Research Council.

Also contributed to:

- RED2022-134947-T Red de Matemática Discreta y Algorítmica (01/06/2023 - 30/09/2025)

## ACTIVITY IN RESEARCH-RELATED EVENTS

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Program Committee Member at ICALP Track B, 2026.

Speaker at FOCS 2025.

Speaker at the CSP World Congress 2024.

Speaker at ICALP 2024.

Speaker at the Birmingham CSP Meeting 2024.

Participant at the CSP World Congress 2023.

Participant at the EPIT 2023 summer school.

Speaker at the BYMAT 2022 Conference

Participant in the 1st, 3rd and 4th editions of the GAPCOMB Workshop.

Part of the organization of the CSL 2020 conference.

Speaker at the Theoretical Computer Science Seminar, University of Oxford.

Speaker at the LIMDA Seminar, Universitat Politècnica de Catalunya.

Speaker at the SCOS Seminar, Graz University of Technology.

## PEER-REVIEW

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SIAM Journal on Computing.

ACM Transactions on Computation Theory.

Symposium on Foundations of Computer Science (FOCS).

Symposium on Logic in Computer Science (LICS).  
European Conference On Combinatorics, Graph Theory And Applications.  
International Colloquium on Automata, Languages and Programming (ICALP).  
Computer Science Logic (CSL).  
Computer-Aided Verification (CAV).  
Verification, Model Checking, and Abstract Interpretation (VMCAI).  
Formal Methods in Computer-Aided Design (FMCAD).  
AAAI Conference on Artificial Intelligence.

## RESEARCH VISITS

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| Technical University of Hamburg<br>Hosted by Antoine Mottet. | <i>23/10/2023 - 28/10/2023</i> |
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## SKILLS

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| <b>Programming languages</b> | C++, Python, Java                                       |
| <b>Software &amp; Tools</b>  | LaTeX, SageMath                                         |
| <b>Languages</b>             | Spanish (native), English (level C1), German (level B1) |